

HAYAT MEDICAL COLLEGE

Histology & Embryology Test II for Year I Medical Students

Date: November 28, 2013

Time allowed: 2:30 hours.

Name: _____

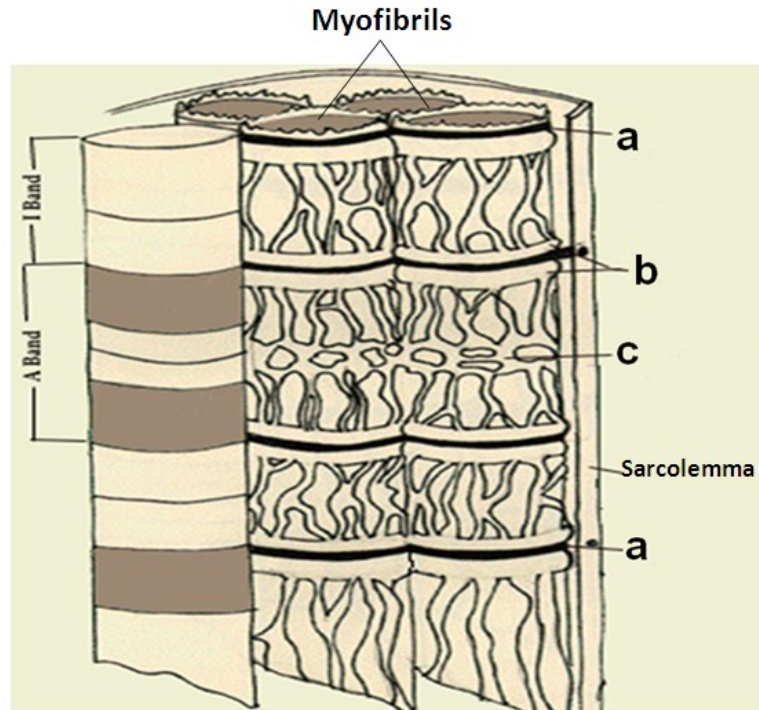
This examination is consisted of _____ pages of question papers and an answer sheet.

Please write your name on both the front page of the examination paper and the answer sheet.

Part I: Histology

For each of the following Multiple Choice Questions, choose the one best appropriate answer and mark "X" in the middle of the appropriate box located against the corresponding question number and letter in the separate answer sheet provided. *(50 points, 1 point for each correct answer)*

1. The sarcoplasmic reticulum of skeletal muscle cells are made by:
a) Golgi apparatus b) Polyribosomes c) Rough endoplasmic reticulum.
d) Smooth endoplasmic reticulum. e) Mitochondria.
2. Regarding the striated muscle cells,
a) The striations of the light microscopy are manifested because of extensive plasma membrane foldings
b) The location of triads in the myofibrils of skeletal muscle cells is identical in all animals.
c) Both skeletal and cardiac muscle cells show regenerative capacity from the surrounding pericytes if damaged.
d) Intercalated discs of cardiac muscle are made by cell junctions.
e) All skeletal muscles are found attached to bones.
- 3-6. Answer questions 8-11 from Figure 1 below representing a muscle cell.



3. Which type of muscle cell does the figure represent?
 - a) Skeletal muscle
 - b) Cardiac muscle
 - c) Smooth muscle
 - d) a & b
 - e) All
4. The regions indicated by letter "a" are:
 - a) Terminal cisternae
 - b) Transverse tubule
 - c) Sarcotubules
 - d) Intercalated duct
 - e) Diad
5. The regions indicated by letter "b" are:
 - a) Terminal cisternae
 - b) Transverse tubule
 - c) Sarcotubules
 - d) Endomysium
 - e) Diad
6. The region indicated by letter "c" is:
 - a) Terminal cisternae
 - b) Transverse tubule
 - c) Sarcotubules
 - d) Mitochondria
 - e) Diad
7. Region of cell bodies in the central nervous system could make all of the following, **except**:
 - a) Gray matter
 - b) Column
 - c) Layer
 - d) Nuclei
 - e) None of the above
8. Nissl bodies of neurons represent:
 - a) Golgi bodies
 - b) Lysosomes
 - c) Microfilaments
 - d) Rough endoplasmic reticulum
 - e) Smooth endoplasmic reticulum
9. Neurofibrils are found in which region of the neurone?
 - a) Dendrites
 - b) Axon
 - c) Perikarya
 - d) a & b
 - e) All of the above
10. Which of the following statements is correct regarding myelination in the peripheral nervous system:
 - a) Outer lamellae end nearest the node of Ranvier in comparison to the inner lamellae.
 - b) The first layer of the myelin sheath known as inner collar of the Schwann cell contains the nucleus.

- c) One Schwann cell makes more than one segment of myelin sheath for same or different axon(s).
- d) Myelin sheath in the peripheral nervous system has fewer Schmidt-Lanterman clefts than that of the central nervous system.
- e) Myelin is made mainly by protein as neurokeratin with a smaller amount of phospholipid.

11. Which of the following statements is **not true** regarding the supporting cells in the nervous tissue:

- a) Satellite cells surround the perikarya of the spinal ganglia in a stratified cuboidal fashion.
- b) Oligodendrocytes are found in both gray and white matters of the central nervous system.
- c) Microglia are the smallest in size among the neuroglial cells.
- d) Astrocytes serve in conducting metabolites between blood vessels and neurones.
- e) Ependymal cells line both the ventricles of the brain as well as the central canal of the spinal cord.

12. Which of the following neuroglia belong to the mononucleated phagocytic cells?

- a) Astrocytes b) Oligodendrocytes c) Microglia d) Ependyma e) a & b

13. Erythrocytes

- a) Stain acidophilic b) Are biconcave c) Have rigid cell membrane
- d) Have centrally placed heterochromatic nucleus e) Are more in number in females than males.

14. The normal value for leukocyte count is

- a) 200,000 – 400,000/ μ l b) 4,100,000 – 6,000,000 / μ l c) 5,000 – 12,000/ μ l
- d) 35,000 - 50,000/ μ l e) 2,000,000-5,000,000/ μ l

15. Which type of leukocytes are least in number in the circulating blood?

- a) Basophils b) Eosinophils c) Monocytes d) Neutrophil e) Lymphocytes

16. Which leukocytes increase most in number during bacterial infection?

- a) Basophils b) Eosinophils c) Monocytes d) Neutrophil e) Lymphocytes

17. At which stage of hematopoiesis do morphological characteristics differentiate for the first time, suggesting the mature cell types they will become?

- a) Multipotential stem cells b) Precursor cells
- c) Progenitor cells d) Pluripotential stem cells e) None of the above

18. Stage of granulopoiesis with twisted or curved nucleus showing uniform width is:

- a) Metamyelocytes b) Myelocytes c) Promyelocytes **d) Myeloblast** **e) Band stage**

19. Which of the following statements is not correct regarding the arteries?

- a) Elastic arteries do not have smooth myocytes in their tunica media**
- b) Tunica media of muscular arteries have elastic and reticular fibres among the smooth myocytes**
- c) Tunica adventitia of muscular arteries have collagen and elastic fibres**
- d) Most of the named arteries are muscular arteries**
- e) Aorta is an example of elastic artery**

20. Which of the following statements is not correct regarding the capillaries?

- a) Continuous capillaries are lined by nonfenestrated endothelium joined by tight junctions**
- b) Fenestrated capillaries are lined by endothelium showing several fenestrations, but have continuous basal lamina**

- c) Hematopoietic organs contain sinusoidal capillaries
- d) Capillaries have thin subendothelial layer
- e) All of the above

21. Which of the following statements is not correct regarding the veins?

- a) Are more in number than the arteries
- b) Have valves which are not found in the arteries
- c) Have thinner wall and larger lumen when compared with the arteries
- d) Have prominent internal elastic lamina
- e) Contain larger volume of blood than the arteries

22. Which of the following is relatively well developed in large veins as compared to those of arteries?

- a) Tunica intima
- b) Tunica media
- c) Tunica adventitia
- d) All of the above
- e) None of the above

23. With regard to impulse generating and conducting system of the heart:

- a) Sinuatrial node is located in the left atrium
- b) Cells of the sinuatrial node are larger in size when compared to those of the ordinary cardiac myocytes
- c) Cells of the sinuatrial node have more myofibrils than those of the ordinary cardiac myocytes
- d) Purkinje cells, when seen under light microscopy following H & E staining, appear as large cells with clear zone surrounding centrally placed nucleus
- e) All are correct

24. Regarding the heart:

- a) Myocardium is thicker in the atria than in the ventricles
- b) Myocardium is made by both cardiac and smooth muscle cells
- c) Some cardiac myocytes act as endocrine cells
- d) Is lined internally by stratified squamous epithelium
- e) Purkinje fibres are found in the epicardium

25. Which of the following statements is true regarding the lymphoid nodules?

- a) Mainly contain T lymphocytes.
- b) Could be found freely in mucosa associated lymphoid tissue in several regions.
- c) Primary lymphoid nodules show germinative region in the centre.
- d) Have connective tissue capsule.
- e) All of the above.

26. Which of the following is **not true** about thymus?

- a) Has six types of epithelial reticular cells.
- b) Attains to its maximum size at puberty.
- c) Contain no B lymphocyte.
- d) Has both primary and secondary lymphoid nodules in its cortex.
- e) Has Hassall's corpuscles preferentially located in the medulla.

27. Which of the following structures **does not** make part of blood-thymus barrier?

- a) Pericytes
- b) Endothelium
- c) Type I epithelial reticular cells
- d) Basal laminae
- e) Smooth myocytes.

28. Which of the following statements is **not true** about epithelial reticular cells in the thymus?

- a) Are stellate cells with processes.
- b) Stain darker than the lymphocytes in haematoxylin & eosin stained preparations.
- c) Isolate cortex from connective tissue & blood vessels.

- d) Separate cortex from medulla. e) All of the above.
29. Which of the following are **not** normally present in the subcapsular sinus of a lymph node?
 a) Macrophages b) Reticular cells c) Lymphocytes
 d) Reticular fibres e) Erythrocytes
30. Which of the following is **not true** about a lymph node?
 a) Afferent lymphatic vessels enter through the hilum.
 b) Efferent lymphatic vessels take away newly formed lymphocytes.
 c) Lymphocytes leave the high endothelial venules in the paracortex.
 d) Has both B and T lymphocytes.
 e) Intermediate lymph sinuses are found in the paracortex.
31. Which of the following statements is **not true** about the spleen?
 a) Is the largest lymphoid organ b) Filters blood.
 c) Has capsule and trabeculae which contain smooth myocytes.
 d) Has no B lymphocytes. e) Has both open and closed blood circulation.
32. Which of the following lymphoid organs is involved in filtering lymph?
 a) Appendix b) Palatine tonsil c) Spleen d) Thymus e) Lymph node

Part II: Embryology

For each of the following Multiple Choice Questions, choose the one best appropriate answer and mark "X" in the middle of the appropriate box located against the corresponding question number and letter in the separate answer sheet provided. (50 points, 1 point for each correct answer)

1. Which of the following statements is **false** regarding the human placenta?
 a) Maternal portion is formed by deciduas parietals.
 b) Fetal portion is formed by chorion frondosum.
 c) Cotyledons are seen from the maternal aspect.
 d) Fetal portion is smooth. e) Is hemochorial
2. Which of the following statements is **false** regarding the human placenta?
 a) The stem villi serve to anchor the placenta onto the maternal tissue.
 b) Floating villi are involved in the exchange of substances between the maternal and fetal blood more than the stem villi.
 c) The fetal blood freely circulates from umbilical arteries into the intervillous space and then back via umbilical vein.
 d) The fetal border is covered by amnion.
 e) At full term its attachment extends to cover about 30% of the endometrium.
3. Which of the following is **false** concerning the amniotic fluid

- a) Normally amounts to 400 - 500 ml during the last trimester.
 - b) About 99% of it is made by water.
 - c) Is produced primarily from the maternal interstitial fluid.
 - d) Is taken into both the gastrointestinal and respiratory tracts of the fetus.
 - e) Contains fetal feces and urine
4. When placenta is developed at the lower segment, near the cervix of the uterus, it is known as:
- a) Placenta previa b) Placenta succenturiata c) Battledore Placenta
 - d) Placenta accrete e) Placenta membranacea
5. When monozygotic twins are formed by split at morula stage, the twins will have
- a) Same chorionic sac and placenta, but separate amniotic cavity
 - b) Same chorionic sac, but separate placenta and amniotic cavity
 - c) Same chorionic sac, placenta and amniotic cavity
 - d) Separate chorionic sac, placenta and amniotic cavity
 - e) Separate chorionic sac and placenta, but same amniotic cavity
6. With regard to the congenital anomalies, which of the following statements is **false**?
- a) At birth, the incidence of minor congenital anomalies is less than that of major congenital anomalies.
 - b) Infants with multiple minor anomalies most frequently have one or more associated major anomalies as well.
 - c) Recessively inheritable mutations may cause birth defect while occurring in single (heterozygous) dose.
 - d) Monosomy of the sex chromosome is known to cause human birth defect seen at birth?
 - e) A similar teratogen may not cause identical congenital anomalies in different genomes.
7. The causes for most of the currently detected major human birth defects at birth are:
- a) Genetic factors b) Environmental factors
 - c) Combination of genetic and environmental factors d) Unknown factors
8. A condition where there is a decrease or an increase from the normal diploid number of a specific chromosome as cause(s) for birth defect is known as:
- a) Polyploidy b) Hypodiploidy c) Aneuploidy d) Hyperdiploidy e) Ploidy
9. Most birth defects occur during which of the following periods/stages of the intrauterine life?
- a) First 2 weeks b) Embryonic period c) Fetal period d) Last trimester e) Zygote
10. Which of the following structures develop from the costal processes of the sclerotomes?
- a) The transverse processes b) Foramina transversaria
 - c) Lateral mass or alae of the sacrum d) All of the above e) None of the above
11. The cranial half of the 1st cervical sclerotome:
- a) Gives rise to the upper part of body of the 1st cervical vertebra.
 - b) Gives rise to the lower part of body of the 1st cervical vertebra.
 - c) Gives rise to the whole body of the 1st cervical vertebra.
 - d) Participates in the formation of base of occipital bone. e) Disappears
12. The body of 1st thoracic vertebra is formed by:
- a) The caudal half of the 8th cervical sclerotome alone.
 - b) The cranial half of the 1st thoracic sclerotome alone.
 - c) The caudal half of the 1st thoracic sclerotome alone.
 - d) Fusion of caudal half of the 1st thoracic with cranial half of the 2nd thoracic sclerotomes.

e) Fusion of caudal half of the 8th cervical with cranial half of the 1st thoracic sclerotomes.

13. The spinal nerves, in relation to their corresponding sclerotomes, were initially located:

- a) Above their corresponding sclerotomes.
- b) Below their corresponding sclerotomes.
- c) At the level of succeeding sclerotome.
- d) At the level of preceding sclerotome.
- e) At the level of middle of corresponding sclerotomes.

14. Bones of the limbs develop from:

- a) Somatic layer of lateral mesoderm.
- b) Visceral layer of lateral mesoderm.
- c) Sclerotomes.
- d) Intermediate mesoderm.
- e) Neural crest.

15. Which of the following statements is correct with regard to the rotation of the limbs during the 7th week?

- a) Lower limb rotates 180° medially, while upper limb rotates 180° laterally.
- b) Lower limb rotates 180° laterally, while upper limb rotates 180° medially.
- c) Lower limb rotates 90° laterally, while upper limb rotates 90° medially.
- d) Lower limb rotates 90° medially, while upper limb rotates 90° laterally.
- e) Lower limb rotates 90° laterally, while upper limb rotates 180° medially.

16. The dorsal somatic mesodermal condensations in the limb bud give rise to which types of muscles?

- a) Extensors and pronators.
- b) Extensors and abductors.
- c) Extensors and flexors.
- d) Flexors and pronators.
- e) Flexors and abductors.

17. Which of the following statements is true regarding the hypomeric and epimeric divisions of the myotomes?

- a) Epimers are larger and located dorsally as compared to the hypomers.
- b) Epimers are smaller and located dorsally as compared to the hypomers.
- c) Epimers are innervated by the dorsal branches of the ventral primary rami of the spinal nerves.
- d) Hypomers give rise to the extensors of the back.
- e) None of the above.

18. Cardiac muscles develop from:

- a) Somatic layer of lateral mesoderm.
- b) Splanchnic layer of lateral mesoderm.
- c) Intermediate mesoderm.
- d) Both splanchnic and somatic mesoderm.
- e) Myotomes.

19. Which of the following is a condition caused by early closure of epiphyseal plates of long bones leading to dwarfism?

- a) Acrania
- b) Acromegally
- c) Achondroplasia
- d) Klippel-Feil syndrome
- e) None of the above

20. Which of the following bones **does not** developed in the 1st arch?

- a) Incus
- b) Malleus
- c) Maxilla
- d) Mandible
- e) Styloid process of temporal bone

21. Structure that is not developed in the third branchial arch includes:

- a) Lesser cornu of hyoid bone
- b) Glossopharyngeal nerve
- c) Common carotid artery
- d) Stylopharyngeus muscle
- e) Lower part of body of hyoid bone

22. Structures derived from endoderm of the fourth pharyngeal pouch include

- a) Thymus gland
- b) Palatine tonsil
- c) Thyroid gland
- d) Para-follicular cells of thyroid gland
- e) Lingual tonsil

23. The anterior 2/3rd of the tongue develops from

- a) Hypobranchial eminence
- b) The median lingual swelling
- c) Coupula

d) The lateral lingual swellings

e) Epiglottal swelling

24. Which of the following **does not** develop from the frontonasal prominence?

a) Primary palate

b) Alveolar process for the premolar teeth of the upper lip

c) Philtrum of the upper lip

d) Forehead

e) Nasal septum

25. Structures derived from endoderm of the second pharyngeal pouch include

a) Thymus gland

b) Palatine tonsil

c) Thyroid gland

d) Parafollicular cells of thyroid gland

e) all of the above

26. Tympanic membrane develops from which germ layer?

a) Ectoderm

b) Mesoderm

c) Endoderm

d) a and c

e) All of the above

27. Which of the following statements is **not correct** regarding to the development of the lymph nodes?

a) Develop from aggregation of mesenchymal cells locally as well as in the lymph sacs

b) Are invaded by the lymphocytes arriving through the lymphatics from external source.

c) Differentiation into cortex and medulla begins during the 2nd trimester

d) During the early stages of development, hemocytoblasts and erythrocytes are present

e) Lymphoid nodules appear before differentiation of cortex and medulla

28. Which of the following is **not** a lymph sac present during the early stages of development of the lymphatic vessels?

a) Jugular lymph sac

b) Iliac lymph sac

c) Retroperitoneal lymph sac

d) Umbilical lymph sac

e) Cisterna chyli

29. Which of the following is **not true** with regard to the early development of the blood and immune system?

a) Leukocytes appear before the appearance of erythrocytes

b) There were two thoracic ducts early in development contrary to the one we have in the adult

c) Erythroblastosis fetalis may be caused when mother is Rh^{-ve} and the fetus is Rh^{+ve}

d) Development of blood cells appear in extraembryonic sites before it appears in intraembryonic sites

e) None of the above

15. Which of the following glands of skin becomes functional during the fetal period?

a) Apocrine sweat glands

b) Merocrine sweat glands

c) Sebaceous glands

d) All of the above

e) None of the above

30. Which of the following statements is **not true** regarding to the development of glands of the skin?

a) Secretory cells develop from ectoderm, while the myoepithelial cells develop from the underlying mesoderm.

b) Apocrine sweat glands develop as side buds of hair follicles superficial to that of the sebaceous glands

c) Mammary glands develop from mammary ridges as bilateral ectodermal thickenings extending from the axillary to the inguinal regions.

d) Mammary glands are developmentally regarded as modified sweat glands.

e) Nipple develops by proliferation of connective tissue in the region of the areola.

31. Which of the following statements is correct regarding to the development of the nails?

a) The nails of both the fingers and toes begin to develop at the same time.

b) The nail plate develops by proliferation and migration of cells from the lateral nail folds.

c) The nail fields initially develop at the tip of the digits, but later migrate onto the dorsal surface of the digits.

d) Fingernails reach the fingertips after birth.

e) None of the above

32. Which of the following is **not** a component of vernix caseosa?

a) Secretions from sebaceous gland

b) Secretions of apocrine sweat glands

c) Fallen epidermal cells

d) Fallen hairs

e) None of the above

33. Which of the following structures in the skin develops from a different germ layer than the others?

a) Hairs

b) Sweat glands

c) Arrector pili muscle

d) Nails

e) Apocrine sweat glands